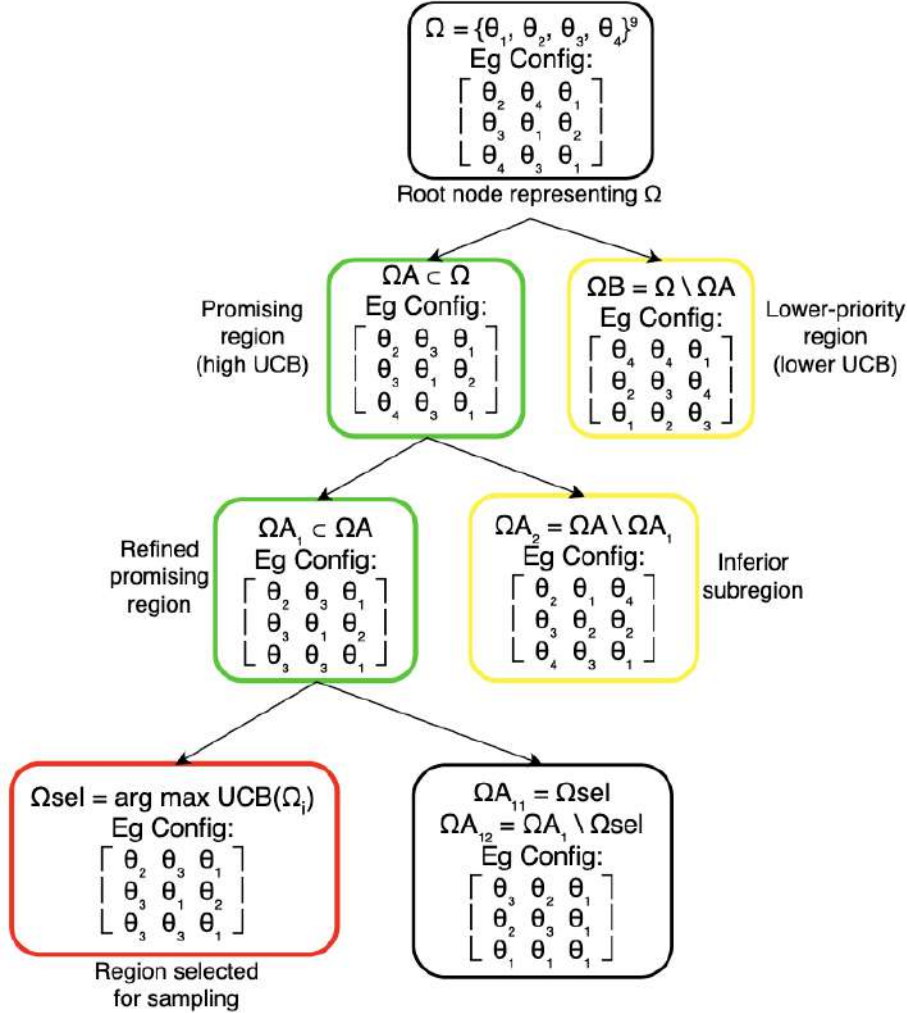


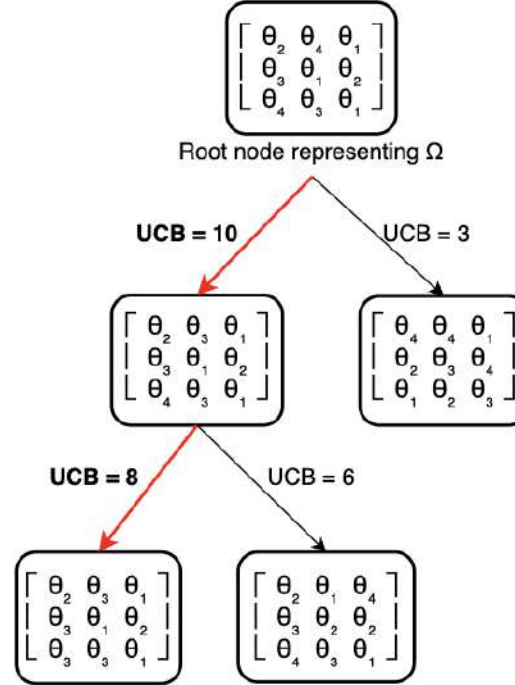
Hierarchical Phase Space Partition*

Each node represents a region, $\Omega \subseteq \{0, \dots, N_{\text{lev}}-1\}^{M \times B}$, where $M \times B$ is the reflection matrix size, and $N_{\text{lev}} = 2^N$

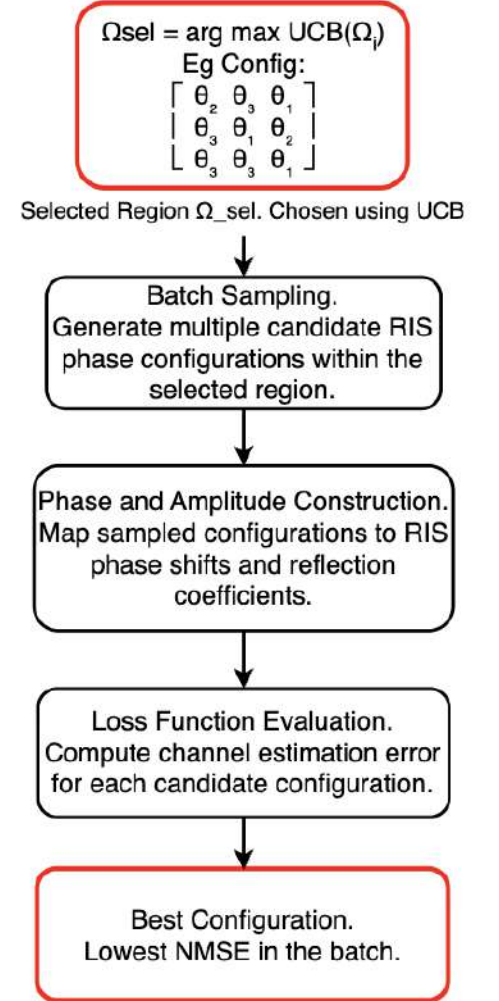


UCB-Guided Region Selection*

Selection: $\arg \max$ UCB over child regions (Representative UCB values)



Batch Sampling and NMSE Evaluation



*The phase matrices shown in each node are representative samples of their regions and do not reflect sequential updates.